

Orientation

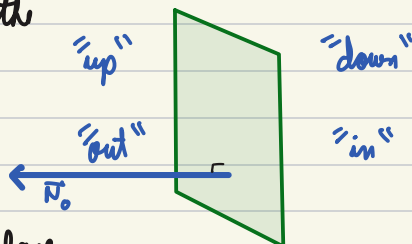
May 6, 2026

How do we describe the orientation of a plane?

We choose a side to be "up" and the opposite side is then "down".

Or, think of the plane as a wall separating the "inside" from the "outside".

We can abbreviate "up/down", "in/out", etc. by choosing a normal vector $\vec{N}_0$ to the plane and declaring the positive orientation to be the side pointing in the same direction as $\vec{N}_0$.



We can use the same vector $\vec{N}_0$ at each point of the plane in order to get a consistent orientation on all the plane.

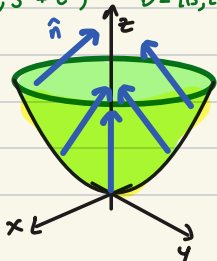
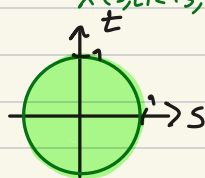
def: Let S be a smooth surface in $\mathbb{R}^3$ (i.e. $S = \vec{X}(D)$ for some parametrized surface $\vec{X}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$, where

the function $\vec{X}$ is one-to-one except possibly on the boundary of D ,
the function $\vec{X}$ is C^1 on all D and $\vec{N}(s,t)$ is never $\vec{0}$ on all of D ,
 $\vec{N}(s,t) \neq \vec{0}$ for all (s,t) in D .

We say the surface S is orientable with orientation given by $\hat{n}$ if there is a continuous function $\hat{n}: S \rightarrow \mathbb{R}^3$ uniquely defining a unit vector $\hat{n}(x,y,z)$ normal to S for each point (x,y,z) in S .

e.g. Can we use a smooth parametrization to give an orientation? } answer

$$\vec{X}(s,t) = (s, t, s^2 + t^2) \quad D = \{(s,t) : s^2 + t^2 \leq 1\}$$



$$\vec{N} = (-2s, -2t, 1)$$

$$\|\vec{N}\| = \sqrt{4(s^2 + t^2) + 1}$$

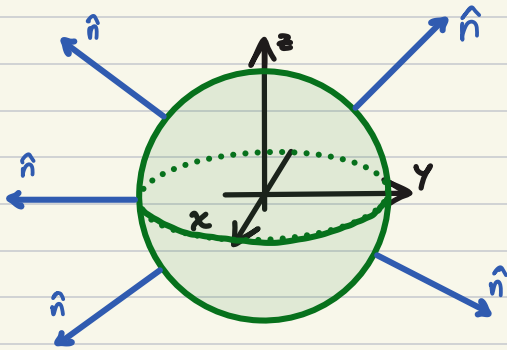
$$\text{define } \hat{n}(\vec{X}(s,t)) = \frac{\vec{N}}{\|\vec{N}\|}(s,t)$$

$$\hat{n}(x,y,z) = \frac{1}{\sqrt{4z+1}}(-2x, -2y, 1)$$

Sometimes, with the appropriate surface and parametrization

Worked here ☺

e.g. Sometimes we can define an orientation without using the parametrization
 S sphere $x^2 + y^2 + z^2 = 1$

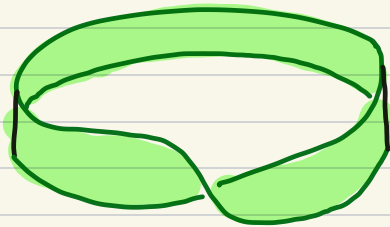
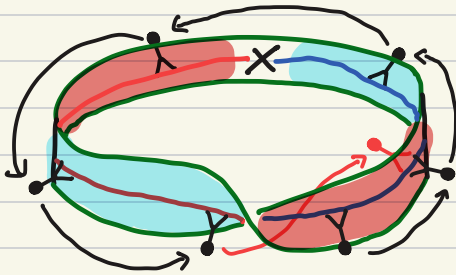


could parametrize by
 $\vec{X}(\theta, \varphi) = (\cos \theta \sin \varphi, \sin \theta \sin \varphi, \cos \varphi)$
 $D = \{(\theta, \varphi) : 0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi\}$

$$\hat{n}(x, y, z) = (x, y, z)$$

Non-orientable surfaces

e.g. Möbius strip

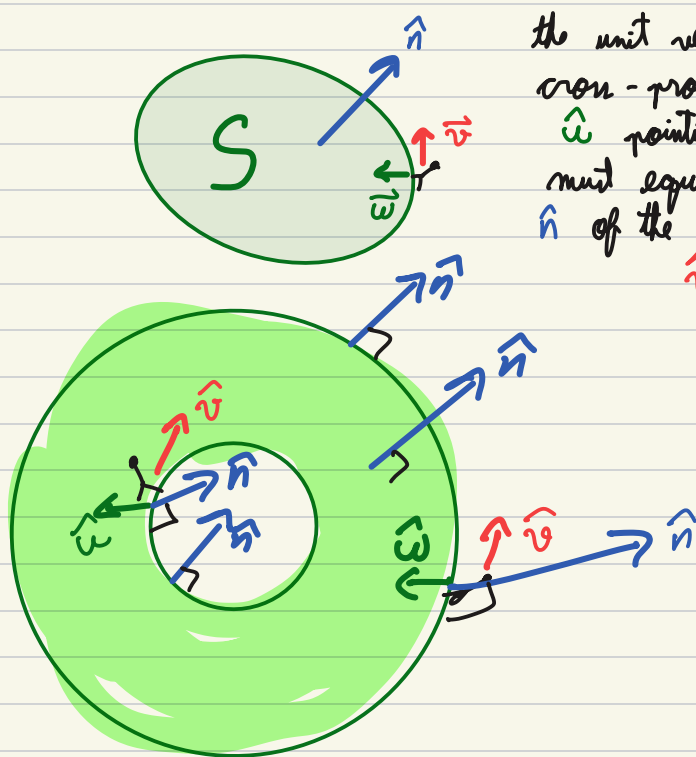


Think of $\hat{n}$ on the figurine λ ,
 can't uniquely and continuously
 draw the figurine.

Smooth orientable surfaces with boundary

Suppose $S = \vec{X}(D)$ is a smooth oriented surface in $\mathbb{R}^3$ with boundary consisting of finitely many piecewise differentiable closed simple curves.

Then the orientation $\hat{n}: S \rightarrow \mathbb{R}^3$ of S induces a compatible orientation on the boundary:



the unit vector $\hat{v}$ of the orientation of ∂S cross-product with the unit vector $\hat{w}$ pointing inside the surface must equal the unit normal vector $\hat{n}$ of the surface orientation

$$\hat{v} \times \hat{w} = \hat{n}$$

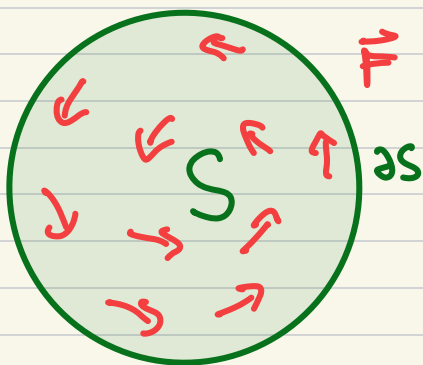
def: cdf S is an oriented surface with orientation $\hat{n}: S \rightarrow \mathbb{R}^3$ and ∂S consists of finitely many closed simple curves, then a compatible orientation for ∂S is the orientation making $\hat{v} \times \hat{w} = \hat{n}$

Stokes's Theorem

Theorem: Let S be a bounded smooth oriented surface in $\mathbb{R}^3$ and suppose the boundary ∂S consists of finitely many piecewise diff. simple closed curves. If ∂S is given an orientation compatible with the orientation of S , then:

For any vector field $\vec{F}$ with domain containing S and being of class C^1 , then

$$\iint_S \nabla \times \vec{F} \cdot d\vec{S} = \oint_{\partial S} \vec{F} \cdot d\vec{s}$$



e.g. Set S be the paraboloid $z = 9 - x^2 - y^2$ defined over the disk $x^2 + y^2 \leq 9$ with orientation pointing "upward" (z -component of $\hat{n}$ positive).

Verify Stokes' Theorem for the v.f.

$$\vec{F} = (2z-y)\hat{i} + (x+z)\hat{j} + (3x-2y)\hat{k}$$

Let's first parametrize the surface and compute the vector surface integral:

Let's try with $\downarrow$ and check the orientation

$$\vec{X}(r, \theta) = (r \cos \theta, r \sin \theta, 9 - r^2) \quad D = \{(r, \theta) : 0 \leq r \leq 3, 0 \leq \theta \leq 2\pi\}$$

$$\vec{T}_r = (\cos \theta, \sin \theta, -2r) \quad \vec{T}_\theta = (-r \sin \theta, r \cos \theta, 0)$$

$$\vec{N} = (2r^2 \cos \theta, 2r^2 \sin \theta, r)$$

z component > 0 (except at $(0,0,9)$) so this orientation coincides.

$$\text{Now } \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2z-y & x+z & 3x-2y \end{vmatrix} = \begin{matrix} (-2-1)\hat{i} + (2-3)\hat{j} \\ + (1+1)\hat{k} \end{matrix} = (-3, -1, 2)$$

And finally the integral

$$\begin{aligned} \iint_S \nabla \times \vec{F} \cdot d\vec{S} &= \int_0^3 \int_0^{2\pi} [-3r^2 \cos \theta - r^2 \sin \theta + 2r] d\theta dr \\ &= \int_0^3 [-2r^2 \sin \theta + 2r^2 \cos \theta + 2r\theta]_{\theta=0}^{\theta=2\pi} dr \\ &= 4\pi \int_0^3 r dr = 2\pi r^2 \Big|_0^3 = \underline{18\pi} \end{aligned}$$

Let's now work on the vector line integral along ∂S :

A sketch of the surface will help us parametrize the boundary with orientation compatible with $\hat{n}$.



$\partial S = \{(x, y, 0) : x^2 + y^2 = 9\}$ oriented by $\vec{v}$ (counterclockwise)

$$\vec{r}(t) = (3 \cos t, 3 \sin t, 0) \quad 0 \leq t \leq 2\pi \quad \vec{v} = \vec{r}' = (-3 \sin t, 3 \cos t, 0)$$

Vector line integral:

$$\begin{aligned} \int_{\partial S} \vec{F} \cdot d\vec{s} &= \int_0^{2\pi} (2(0) - 3 \sin t, 3 \cos t + 0, 9 \cos t - 6 \sin t) \cdot (-3 \sin t, 3 \cos t, 0) dt \\ &= \int_0^{2\pi} 9 \cos^2 t + 9 \sin^2 t dt = \int_0^{2\pi} 9 dt = 18\pi \end{aligned}$$